

AKSHAR VANDARA

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EDUCATION

Northeastern University, Boston, MA	May 2026
<i>Master of Science, Game Science and Design</i> , GPA 3.87/4	
Relevant Courses: Game Data Science, Mixed Research Methods in Games, Generative Game Design	
Gujarat Technological University, Ahmedabad, India	Jun 2023
<i>Bachelor of Engineering, Computer Engineering</i> , GPA 3.6/4	
Relevant Courses: Operating System, Software Engineering, Artificial Intelligence, Cloud Computing	

TECHNICAL SKILLS

- **Game Dev:** Unity, Unreal Engine, Blueprints, Game Physics, Core-gameplay management, Database handling
- **Game Design:** Animations, Animation Rigging, UI integration, Environment Design
- **Programming Languages:** C#, C++, JavaScript, Python, Java, R, C, SQL
- **Python Libraries:** NumPy, Pandas, PyTorch, Cuda, TensorFlow, PyGames, Flask, Django, SQLAlchemy
- **Front-End:** HTML, CSS, JavaScript, Three.js, Tailwind, React, Node, Vue, Angular, Next.js, Vite.js
- **Developer Tools:** Git, Perforce, Firebase Realtime DB, Firebase Cloud functions, SourceTree, Gitlab, Figma, Trello

PROFESSIONAL EXPERIENCE

Research Assistant

Northeastern University, Boston, MA, USA	Feb 2025 – Present
• Extended the 'Sturgeon' constraint-solving system to generate 2D puzzle levels along with their playthroughs. • Improved the generation algorithm's performance by over 100x and increased its reliability over prior academic methods. • Co-authored a paper detailing the system's architecture and results, published at the AAAI 2025 AIIDE conference. • Currently working on generating 3D game levels by applying patterns extracted from 2D games to the system.	

Unreal Developer

Otisco Studios, FL, USA	May 2025 – Aug 2025
• Developed 8+ core gameplay systems for two titles, covering save management, UI, and environment mechanics. • Implemented AI for enemies and NPCs, delivering 5+ archetypes in roguelike and open-world genres using Behavior Trees and Blackboards. • Identified and fixed over 30+ bugs prior to launch, directly improving product stability. • Profiled and refactored code across multiple projects, increasing average FPS by ~100% during gameplay.	

Unity Developer

Arcadon Games, Bangalore, India	Mar 2023 – Jul 2024
• Engineered core mechanics for Cricket Tycoon, using state machines and player interaction systems, achieving a 30% improvement in gameplay responsiveness. • Integrated Firebase backend solutions, managing over 1,000+ user data entries, inventory systems, and game progress synchronization with real-time updates via cloud functions. • Designed AI simulation systems capable of simulating up to 30 randomized matches per minute, enhancing gameplay variety and reducing load times by 40%. • Conducted alpha testing with 100+ users, achieving 85% positive feedback, and resolved 20+ critical issues to significantly improve game stability and user engagement.	

PROJECTS

Spacetime Level Generation Link	Feb 2025 – Present
• Modified a Python constraint-solving system ('Sturgeon') to generate 2D puzzle game levels along with their solutions. • Increased level generation speed by 100-500 times and improved error-proofing compared to previous methods like STWFC (Space-Time WaveFunctionCollapse) • Co-authored and published a research paper on the system that was accepted into the 2025 AAAI conference (AIIDE).	

Red Circuit Rebellion Link	Dec 2024 – April 2025
• Developed 5+ core gameplay systems, including save/load, environmental fracturing, health management, and objectives manager. • Implemented enemy AI using Behavior Trees and EQS to create 3 combat archetypes. • Built 2 game modes (Story and Arcade) with specific objective logic and enemy spawners for each. • Optimized game assets and blueprints, increasing average FPS by ~100% during gameplay.	

Cricket Manager: League Link	Mar 2023 – Jul 2024
• Built sports management system capable of handling 1000+ concurrent users with Firebase, maintaining sub-100ms response times for pseudo-multiplayer features. • Orchestrated alpha testing phase with 100+ users, iterating through 3 major versions to achieve 85% user satisfaction and 40% reduction in bug reports.	